

American International University-Bangladesh (AIUB)
Faculty of Science and Technology

AI-Based Dynamic Traffic Signal Control System With Emergency Vehicle Prioritization for Urban Intersections in Bangladesh.

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Declaration

This thesis is composed of our original work, and contains no material previously published or writtenby another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financialand any other original research work used or reported in our thesis. The content of our thesisis the result of work we have carried out since the commencement of the Thesis.

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Author Contributions

List the significant and substantial inputs made by different authors to this research, work and writingrepresented and/or reported in the thesis. These could include significant contributions to the conception and design of the project; non-routine technical work; analysis and interpretation of researchdata; drafting significant parts of the work or critically revising it to contribute to the interpretation.

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0 - 3 points	**Perform as effective individual**			
Critical thinking				
Reflection on feedback				
Quality of work				
Self-directed				
0 - 3 points	**Perform as effective team member/leader**			
Taking responsibility				
Contribution				
Collaboration				
Working with others				
0 - 3 points	**Perform as effective team member/leader**			
Presentation delivery				
Voice and tone				
Enthusiasm				
Creativity & Tools use				

Project-Thesis Planning

Table 01: Project-Thesis Deliverables

Project Tasks	Start Date	End Date	Execution Data
1. Project Planning	2025.07.07	2025.07.15	Meet supervisor, refine objectives, define scope, deliverables and milestone plan.
2. Literature Review	2025.07.16	2025.07.24	Collect key papers, summarize methods, datasets, gaps and define baseline approaches.
3. Dataset Preparation	2025.07.25	2025.08.10	Collect image for dataset, preprocessing : resize/augmentation, and create train/valid/test splits.
4. Baseline Model Analysis	2025.08.11	2025.08.18	Implement baseline YOLOv5 model, set training loop and log initial metrices.
5. Model Training	2025.08.19	2025.08.31	Run model training and get accuracy metrices.
6. Dynamic Timing	2025.09.01	2025.09.10	Fix dynamic timing logic, implement with the object detection code and check for accuracy.
7. Four-Lane Model Implementation	2025.09.11	2025.09.22	Take four-lane footage, implementation in the baseline model.
8. Experiments, Tuning and Evaluation	2025.09.23	2025.09.25	Run experiments, compare with baseline, generate metrices and plots.
9. Thesis Writing	2025.09.26	2025.10.03	Write methodology, experiments, results and discussion, insert table, figures and graphs.
10. Final Draft Preparation	2025.10.04	2025.10.07	Fix formatting, finalize references, prepare slides and submit thesis.

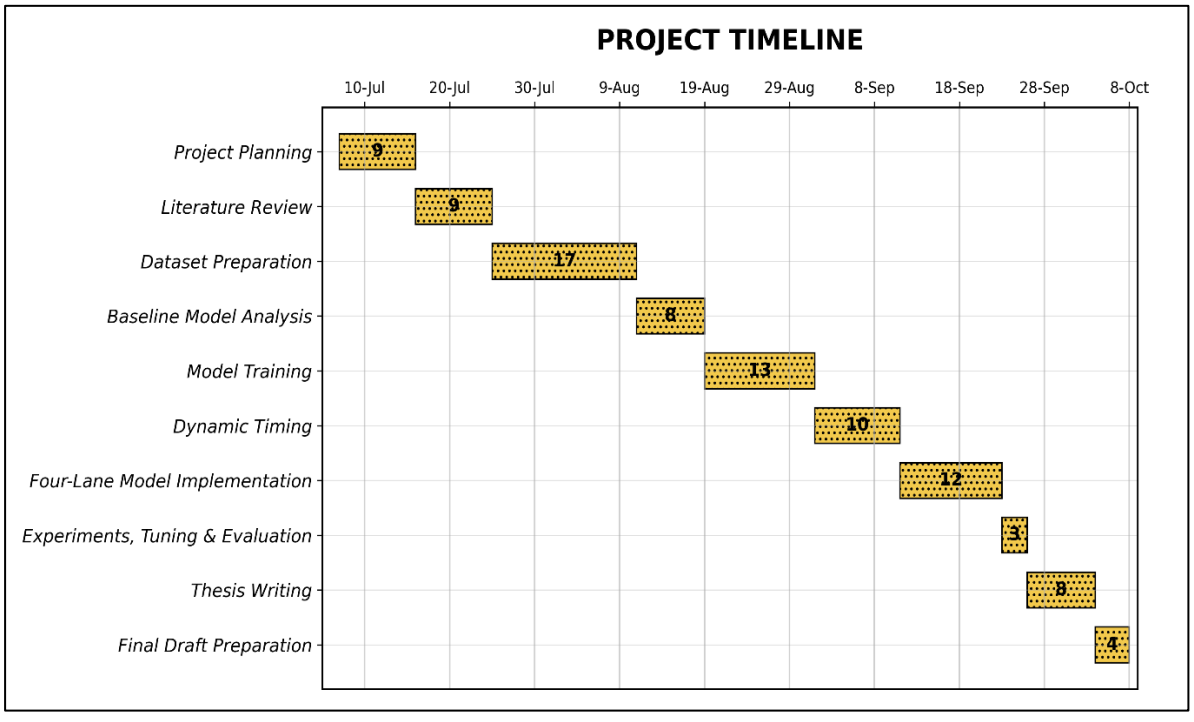


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List of Abbreviations

AI:	Artificial Intelligence
API:	Application Programming Interface
ARI:	Accident Research Institute
BIDS:	Natural Language Processing
BRTA:	Bangladesh Road Transport Authority
BUET:	Bangladesh University of Engineering and Technology
CCTV:	Closed Circuit Television
CNG:	Compressed Natural Gas
CSE:	Computer Science and Engineering
DMP:	Dhaka Metropolitan Police
DSRM:	Design Science Research Methodology
EMV:	Emergency Vehicle
EMV-TPC:	Emergency Vehicle Traffic Priority Control
EMV-TTP:	Emergency Vehicle Travel Time Prediction
FIFO:	First in First Out
FST:	Faculty of Science and Technology
GDP:	Gross Domestic Product
GPU:	Graphics Processing Unit
HUD:	Heads Up Display
IDSS:	Informal Decision Support System
ITS:	Intelligent Transportation System
LIDAR:	Light Detection and Ranging
mAP:	Mean Average Precision
PANet:	Path Aggregation Network
PC:	Priority Control
ROI:	Regions of Interest
RQ:	Research Question
SSP:	Spatial Pyramid Pooling
SUV:	Sport Utility Vehicle
SV:	Social Vehicle
UAV:	Unmanned Aerial Vehicle

USA:	United States of America
VIP:	Very Important Person
VGG16	Visual Geometry Group 16 (a deep learning model)
VANET:	Vehicular Ad-Hoc Network
WBS:	Work Breakdown Structure
YOLO:	You Only Look Once
YOLOv8m:	You Only Look Once Version 8 Medium
YOLOv8n:	You Only Look Once Version 8 Nano
YOLOv9:	You Only Look Once Version 9

Abstract

Traffic congestion in Dhaka and other major Bangladeshi cities severely delays emergency services, where ambulances and fire trucks often remain stuck at intersections controlled by fixed-time signals that cannot sense real-time demand or provide priority [11]. The thesis introduces a traffic signal control system that is AI-based, dynamic, and urban intersection specific to Bangladesh. The system integrates real-time multi-class vehicle detection, adaptive signal timing, and explicit emergency vehicle prioritization. A YOLOv5 object-detection model serves as the backbone of the system, and the model is trained on a 22-class dataset that is specific to Bangladesh and comprises primary images taken by the authors including dedicated shots of fire trucks taken in collaboration with the Bangladesh Fire Service and Civil Defence along with secondary images from the public "Traffic Bangladesh" dataset [19]. The processing of all images is done by resizing them to 416×416 and performing geometric and photometric transformations to improve generalization through augmentation.

The YOLOv5 model that has been trained gives detections for every frame which then gets translated into vehicle counts for each lane and emergency vehicle flags. These results are then used as input for a lightweight controller, which is defined mathematically, that calculates adaptive green and red times from a weighted sum of current density, previous density, and batch size. In the absence of an emergency vehicle, the controller runs in "Normal" Mode, where it reallocates green time to the more congested approaches while maintaining a fixed cycle length. In case of a detected ambulance or fire truck, the system changes to "Emergency" Mode, giving the priority green to the corresponding approach via direction-based preemption and then safely transitioning back to normal operation. For the real-world validation, the entire process is implemented in a four-arm intersection that is monitored with a single 360-degree camera being the only one used, which is logically divided into four areas of interest that represent the North, East, South, and West approaches.

The experimental results obtained from the held-out test set illustrate that the new model scores an impressive 70.0% in Precision, 71.6% in Recall, 70.8% in F1-Score, mAP@0.5 of 0.731 and mAP@0.5:0.95 of 0.442. When compared to the baseline model that was only trained using the secondary dataset (mAP@0.5 55.4%, Precision 66.4%, Recall 50.4%), the proposed model gets 17.7 percentage points in mAP@0.5 and 21.2 points in Recall as absolute gains, which clearly indicates the importance of the built dataset and the preprocessing pipeline. The qualitative analyses of confusion matrices, training curves, and four-lane intersection scenarios back up the claim that the system is able to detect emergency vehicles in a reliable manner, lead the traffic signal timing to dynamically adapt, and serve as a feasible foundation for cutting emergency response times and boosting the overall intersection efficiency in Bangladeshi cities.

Keywords

intelligent traffic signal control, yolov5, emergency vehicle priority, computer vision, bangladesh traffic, deep learning, 360-degree camera, adaptive signal timing.

CHAPTER 1

INTRODUCTION

1.1 Background Analysis

Traffic congestion has become one of the most important urban issues in Bangladesh, especially in densely populated cities like Dhaka and Chattogram. The root causes behind this persistent gridlock include: A chaotic mix of vehicle types such as buses, motorbikes, rickshaws, pedestrians competing for narrow road space without lane discipline or reliable signal coordination, poor road infrastructure, reliance on static and fixed-time traffic signals that ignore real-time traffic conditions and fail to adapt when congestion fluctuates, lack of priority mechanisms for emergency responders**,** causing critical delays and risking lives in urgent situations and lack of real-time management strategies have worsened the situation[1]-[4]. The severity of this situation in Dhaka is well documented. Average vehicular speed in the city has fallen to less than 4.8 km/h, down from around 21 km/h about a decade ago, placing Dhaka among the slowest-moving cities in the world [5]. Daily gridlocks are estimated to cost the economy billions of dollars and deprive citizens of roughly 2.4 million working hours each day [5]. According to BUET's Accident Research Institute (ARI), congestion in Dhaka leads to an annual economic loss of approximately USD 6.5 billion, primarily through lost productivity, fuel wastage, and congestion-induced health costs, and indirectly reduces the national GDP by nearly 10 percent every year. A separate analysis by the Bangladesh Institute of Development Studies (BIDS) reports that daily traffic jams consume around 5 million working hours, indicating the scale at which mobility constraints are undermining economic and social well-being. "If ambulances did not face as much traffic as they do every day, about 200 people could have been saved daily in Dhaka city," estimates Golam Mustafa, president of Bangladesh Ambulance Owners Association. [13].

The situation is particularly critical for emergency services. Ambulances, fire trucks, and police vehicles frequently become trapped in long queues at intersections because existing signal systems cannot detect them or provide any form of priority. Delays of just a few minutes can result in preventable fatalities or severe property damage.

Advances in computer vision and artificial intelligence (AI) now make it possible to analyze live CCTV or 360-degree camera feeds and control traffic lights dynamically. In this context, deep learning–based object detection models such as YOLOv5 can detect and classify multiple vehicle types in real time with good accuracy. Building on this capability, this thesis proposes an AI-based dynamic traffic signal control system that uses YOLOv5 to detect vehicles from live video, estimate lane-wise vehicle density, and adapt green phases accordingly, while giving explicit priority to ambulances and fire trucks at urban intersections in Bangladesh.

1.2 Existing Studies

Traffic signal systems and vehicle detection in urban environments have been studied for many years, both in Bangladesh and abroad. One of the earlier and most relevant works for the Dhaka context is by Rahman and Khadem [6]. They systematically analysed the existing traffic signal system in Dhaka and found that the majority of intersections operate with rigid, fixed-time signal plans that are rarely updated. Because these plans do not reflect real-time traffic demand, green time is often wasted on lightly used approaches while heavily loaded arms experience long queues. The authors reported that this mismatch between signal timing and actual flow leads to excessive and unnecessary waiting time for daily commuters, office workers, and students, and contributes to the public perception that traffic signals are ineffective or even counter-productive [6].

Building on the recognition that static timing is a major weakness, Islam et al. proposed a sectored, human-centric intelligent traffic signal control system for Dhaka [7]. In their approach, each intersection is divided into multiple "sectors" or zones, and AI-based logic is used to estimate vehicle density in each zone and adjust the signal timing accordingly. The system is designed to be operated through a user-friendly dashboard so that traffic police and operators can monitor real-time conditions and intervene when needed. Islam et al. also emphasize security and reliability: their architecture uses secure APIs and a centralized platform to reduce the risk of external tampering and cyber-attacks. However, while the system clearly moves beyond traditional fixed-time control, it focuses mainly on adaptive timing and system design. It does not provide a fully integrated, vision-based emergency vehicle priority mechanism, and it pays limited attention to unauthorized or abnormal traffic behaviour, such as rule-breaking three-wheelers or non-motorized vehicles using restricted lanes [7].

A different line of work is presented by Aworinde et al., who developed a prioritized traffic light control system specifically for emergency vehicles [8]. Their system listens for siren sounds from ambulances or fire trucks and, when a siren is detected, automatically changes the traffic signal to provide a green phase for the approaching emergency vehicle. This audio-based strategy allows emergency vehicles to clear intersections more quickly without relying on manual intervention from traffic police. The study demonstrates that even relatively simple sensing (siren detection) can significantly reduce delay for emergency services. However, the proposed design does not incorporate modern AI-based visual detection models such as YOLO, and it gives less attention to maintaining smooth flow for ordinary traffic when no emergency vehicle is present. As a result, it does not fully exploit the potential of combined vision- and audio-based control or address broader traffic optimization beyond emergency priority [8].

To address the need for accurate, real-time vehicle perception in congested Bangladeshi conditions, Fahim fine-tuned a state-of-the-art YOLOv9 model on the locally curated Poribohon-BD dataset [9]. This dataset contains 15 classes of native Bangladeshi vehicles, capturing the heterogeneous traffic mix found in Dhaka. The fine-tuned model achieved an impressive mAP@0.5 of 0.934, outperforming earlier YOLO-based attempts on regional datasets [9]. Beyond reporting strong detection accuracy, Fahim proposed using the model's

outputs to build a spatiotemporal graph of vehicle movements across a CCTV network, enabling high-level applications such as flow prediction, congestion analysis, incident detection, emission estimation, and accident risk prediction. This work clearly shows that deep learning models, when trained on region-specific data, can handle dense and complex traffic scenes similar to those in Dhaka and can feed into broader Intelligent Transportation System (ITS) applications. However, the focus is primarily on detection and analytics; the study does not directly link detection results to real-time signal control or emergency vehicle prioritization.

Complementing this, Nabil et al. concentrated on a very specific but important safety issue: the real-time detection of three-wheelers on expressways in Bangladesh [10]. Since three-wheelers such as auto-rickshaws and easy-bikes are typically banned from expressways for safety reasons, automatic detection can help enforcement agencies prevent them from entering restricted zones. Nabil et al. collected a custom dataset of about 700 images from crowded roads in Rajshahi and trained a YOLOv8-medium (YOLOv8m) model, achieving a mean Average Precision of 88.2%, compared to 83.6% for the lighter YOLOv8-nano variant [10]. They also integrated ByteTrack to track vehicles and count them over time, enabling on-the-ground deployment at checkpoints. Their results highlight the trade-off between accuracy and speed when choosing model sizes and demonstrate how region-specific datasets and tailored architectures can improve enforcement and safety on Bangladeshi expressways [10].

1.3 Research Gap

Based on the above, there is still no integrated, real-time traffic management system for Bangladesh that:

(i) detects a wide range of vehicle types from live video feeds.
(ii) dynamically allocates green time according to lane-wise vehicle density to minimize overall delay.
(iii) prioritizes emergency vehicle movements through a principled, algorithmic preemption policy at intersections.

1.4 Research Motivation and Objective

Motivation:

Recent incidents in Dhaka have clearly shown how traffic congestion can turn into a life-threatening barrier for emergency response. On 25 November 2025, a massive fire broke out at the Korail slum. According to Dhaka Post, 11 Fire Service units were dispatched to the scene, but none of them could reach the affected area even after more than 30 minutes because surrounding access roads were completely blocked by stand-still traffic [11]. Firefighters reported severe difficulty in moving their vehicles through the congested corridors, and local residents had to begin fighting the fire themselves while waiting for professional support to arrive [11]. A similar situation occurred on 17 August 2025, when a devastating blaze erupted at a filling station in Mohakhali. As reported by Bangladesh Pratidin, multiple fire trucks faced extreme difficulty reaching the site; some vehicles were reportedly stuck for about 27 minutes in congestion before additional units could get close enough to intervene [12]. In the context of a fuel-station fire, where every minute increases the risk of explosion and widespread damage, this kind of delay illustrates how dangerous the current traffic environment can be for both responders and the public [12]. Beyond these individual events, frontline stakeholders have warned that this is a daily, systemic problem, not an occasional exception. In an in-depth report by The Business Standard on the need for dedicated ambulance lanes in Dhaka, Golam Mustafa, president of the Bangladesh Ambulance Owners Association, estimated that if ambulances did not face such extreme congestion, nearly two hundred lives could be saved every day in the city [13]. His statement underlines that intersection delays are not just an inconvenience; they translate directly into avoidable deaths and long-term health complications for critical patients who cannot reach hospitals in time [13].

These cases are common and they expose a systemic limitation of static-time signals that ignore real-time lane demand and offer no emergency vehicle prioritization. Motivated by these gaps, this paper combines real-time vehicle detection using YOLOv5 with dynamic signal timing and prioritize ambulance and fire truck to reduce waiting time while preserving fairness for non-priority traffic.

Objectives:

1. Build a real time vehicle detection module using YOLOv5 tailored to Bangladeshi vehicles which consists of 22 classes.

2. Design a dynamic signal control system that sets traffic lights from lane-wise vehicle density.

3. Implement emergency vehicle priority feature that clears the earliest arriving emergency vehicle lane and guarantees rapid service with automatic return to normal operation.

1.5 Research Questions (RQs)

RQ1: Can YOLOv5 accurately detect emergency vehicles in real-time at intersections?

RQ2: Can traffic signals be dynamically controlled based on this model?

RQ3: How much response time improvement can be achieved?

RQ4: Who will be benefited from this solution?

1.6 Research Contribution

The proposed system aims to enhance traffic operations and public safety in Bangladesh. More specifically, the system directly enhances the functionality of the Civil Defence and Fire Service crews, the Traffic Division of DMP, and the traffic signal operations of the Dhaka city. This is accomplished by transitioning from current practices that employ static signal plans and the manual control of signals to a more responsive traffic management system based on current conditions. The implementation of a dynamic, priority- based Traffic Management system will allow the arrival of Emergency Response vehicles (i.e., Ambulances & Fire Trucks) to their destinations significantly quicker and with greater reliability. Additionally, when emergency conditions do not exist, other road users (i.e., Commuters, Students & Pedestrians) will experience less waiting due to longer green signal periods driven by lane-by-lane volume demands instead of pre-determined times. Therefore, the proposed system will not only provide a viable emergency response solution, but it will also enhance the operational efficiency of intersections within Dhaka & Chattogram during regular operating hours.This thesis provides an integrated perception-to-control loop for a large number of traffic in Bangladesh at the research and engineering level. A YOLOv5-based detector customized for 22 local classes feeds a lightweight, explainable controller that transforms lane-wise density into adaptive green splits and prioritized clearance for fire trucks and ambulances. The focus of the work is on deployability on modest compute, transparent decisionmaking, and lowlatency execution. The full pipeline is implemented with an emphasis on deployability on modest computing hardware and low-latency execution, making it realistic for use in existing signal operation centres.

Overall, the main contribution of this work is to show how modern deep-learning-based perception can be tightly coupled with simple, interpretable control logic to address two long-standing problems at Bangladeshi intersections: inefficient fixed-time signalling and the absence of systematic emergency vehicle priority. By demonstrating a working prototype for a four-lane intersection and evaluating it using standard detection metrics and scenario-based analysis, the thesis offers a concrete foundation that can be extended in future with features such as audio-based siren recognition, multi-camera fusion, or coordination across multiple intersections as part of broader smart-city traffic management initiatives.

CHAPTER 2

LITERATURE REVIEW

In the context of Bangladesh, Islam et al. (2024) proposed a secured, intelligent traffic signal system tailored to the chaotic, non-lane-based transportation environment of Dhaka. Their approach includes real-time signal adaptation using AI, with special provisions for emergency vehicle detection and VIP movement. The system is designed to be human-centric, incorporating a user-friendly dashboard and secure APIs to prevent external tampering. The authors reported ongoing real-world experimentation at BUET Palashi Mor and planned to expand trials to other key intersections in Dhaka. Notably, the study highlights the critical need for integrating cybersecurity measures in AI-based traffic systems—a dimension often overlooked in existing literature. Their model directly addresses the Dhaka context, which features a heterogeneous mix of vehicles and unpredictable pedestrian movement[1].

To investigate interactions between motorized and non-motorized vehicles, particularly small road users, such as motorcycles or bicycles, Ansariyar and Jeihani (2023) implemented LIDAR sensor technology to record vehicle-bicycle conflicts at intersections. While their research centered on Baltimore, USA, their findings hold significance for Bangladesh, where motorcycles make up most of the traffic mix but are under-monitored at intersections. Their vehicle-bicycle conflict monitor employed real-time data collection to track speed, direction, and other potential conflict points for both vehicles and cyclists using 3D LIDAR and image processing. The authors concluded that LIDAR is able to provide an objective means of identifying near-crash scenarios and minimizing crash risk for vulnerable road users. For Bangladesh, the ability to adapt this study may help to diminish the research gap associated with identification and safety monitoring of motorbikes and help to address poor standard CCTV limitations associated with low-light and poor weather conditions[2].

Fahim (2024) provides a comprehensive study of the application of deep learning for vehicle detection in Intelligent Transportation Systems (ITS) in Dhaka, Bangladesh. The author fine-tuned the state-of-the-art YOLOv9 model on the locally curated Poribohon-BD dataset, which included 15 classes of native Bangladeshi vehicles. The model achieved a mean Average Precision (mAP@0.5) value of 0.934, outperforming prior YOLO-based attempts on regional datasets. In illustrating the effectiveness of YOLOv9 in utilizing dataset to handle varying and congested traffic, this paper proposes a conceptual framework to implement this model on city-wide networks of CCTV cameras. The output data is organized in the form of a spatiotemporal graph to implement high-levels of traffic data analysis such as flow prediction, congestion analysis and incident extraction. Additionally, the author expands to other ITS applications related to emission estimation, accident risk prediction, and policy evidence, thus bridging a gap between AI research and real-world solutions to urban mobility. This research emphasized the importance of tuning the model to the region, with a practical example of how AI for traffic management can be implemented on a large scale in developing megacities[3].

Nabil et al. (2024) discuss an important problem in the setting of intelligent transportation systems (ITS) with regard to the real-time detection of three-wheelers (e.g., auto-rickshaws, easy-bikes) regulation on expressways, as they are typically prohibited due to safety issues for both operators and other road users. The authors use a weakly supervised source employing YOLOv8 and its object detection capabilities to produce a system for real-time identifying three-wheelers being used by roadway users on expressways, using CCTV. The authors

collected a custom dataset of 700 images from crowded urban roadways in Rajshahi, and augmented and annotated it for improved robustness. By training the YOLOv8 medium (YOLOv8m) version of YOLO, the authors achieved a mean average precision (mAP) of 88.2%, compared to the nanon YOLOv8n version, which achieved a mAP of 83.6%. The authors also used ByteTrack so they could track objects, as well as, for supervision vehicle counting, which successfully allows for on-the-ground deployments at police checkpoints, where they can prevent three-wheelers to illegally enter expressways. The authors mention the trade-off between accuracy and speed, which determined their decision to use the medium size model instead of the more accurate large version, and improved processing speed and reduced latency, while keeping a performance threshold for detection reliability. Overall, this article highlights the key need for regionally specific vehicle detection models, and demonstrates the impact deep learning can have in improving the safety of expressways in developing countries[4].

Devika et al. (2024) proposed an AI-based approach for controlling traffic signals that combined adaptive signaling with emergency vehicle priority in order to improve traffic flow and safety in urban areas. The proposed system also used the YOLOv8 deep learning architecture for real-time detection and classification of vehicles in conjunction with a VGG16 based siren recognition model for the accurate identification of emergency situations. The authors identified that more advanced systems did not use fixed-timer traffic signals and proposed a new model that was based on weighted algorithm which used both the number of vehicles in the current and previous intervals. The proposed system also used fiber optic communication to link signals and provide coordinated traffic flow through intersections. Evaluations from simulations showed that the system reduced delay, congestion, and provide emergency vehicles immediate access by changing signals to green when it detected possible emergency vehicles. The authors indicated that the presented approach was a scalable and able to adapt on their own for intelligent transport systems. Besides improving safe and smooth traffic flow in urban areas, the proposed solution assists with the inherent inefficiencies of static timers[5].

Hao et al. (2024) provide a systematic and thorough literature review of methods of optimizing routes and enabling priority control (PC) for urban emergency vehicles (EMVs). The authors emphasize that the primary contribution to public safety and urban sustainability hinges on minimizing the response times of EMVs; this review presents a systematic and comprehensive assessment of the current status and development of EMV routing optimization and PC for the future. A bibliometric analysis was conducted for trending topics in EMV routing and PC research using VOSviewer, and organized into three areas of focus: EMV Travel Time Prediction (EMV-TTP), EMV Routing Optimization, and Emergency Vehicle Traffic Priority Control (EMV-TPC). The authors note a few limitations of existing reviews (e.g., not accounting for the differences and similarities between EMVs and social vehicles (SVs)); the lack of consideration of EMV-TTP in existing reviews; and lack of consideration of road-link strategies or emerging technologies for EMVs-treatments of all types of Emergency Vehicles (as defined by the Toronto City bylaws). The review of emergent methodologies was classified into model based, simulation based, API based, and UAV based. Further, the authors identified EMV routing PC specific research gaps and areas for future research, such as: using actual EMV data from emergency response bodies; including relevant characteristics of EMVs into models; developing active PC programs and policies; minimizing adverse impact on SVs when establishing optimte EMV routes; and increasing the adoption of emerging methodologies (e.g., UAVs, and VANETs). Overall, the review serves as a valuable starting point for

researchers and practitioers through acknowledgment of interdisciplinary EMV optimization. The opportunities for improving the development of intelligent, efficient, and equitable emergency response ssytems in a smart city context [6].

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Conceptual Framework

This research adopts a Design Science Research Methodology (DSRM) to build, demonstrate, and evaluate an AI-based traffic signal control system with emergency vehicle prioritization. The approach unifies perception (object detection) and control (dynamic signal timing + preemption) in a real-time loop.

DSRM Phases & Mapping to Our Work:

1. **Problem Identification & Motivation**: Fixed-time signals in Dhaka ignore real-time demand and cause critical delays for ambulances/fire trucks.
2. **Define Objectives of the Solution**: (i) robust, real-time vehicle detection across 22 classes; (ii) dynamic green-time allocation from lane-wise density; (iii) preemption for emergency vehicles with safe recovery to normal operation.
3. **Design & Development**: YOLOv5-based perception, lane assignment for 4-lane intersection, density estimation, and a timing controller with emergency logic.
4. **Demonstration**: End-to-end pipeline on recorded intersection video; visual HUD and annotated output video for operator audit.
5. **Evaluation**: Perception metrics (Precision, Recall, F1, mAP@0.5, mAP@0.5:0.95), and operational checks (e.g., change in emergency waiting time, qualitative fairness across lanes).

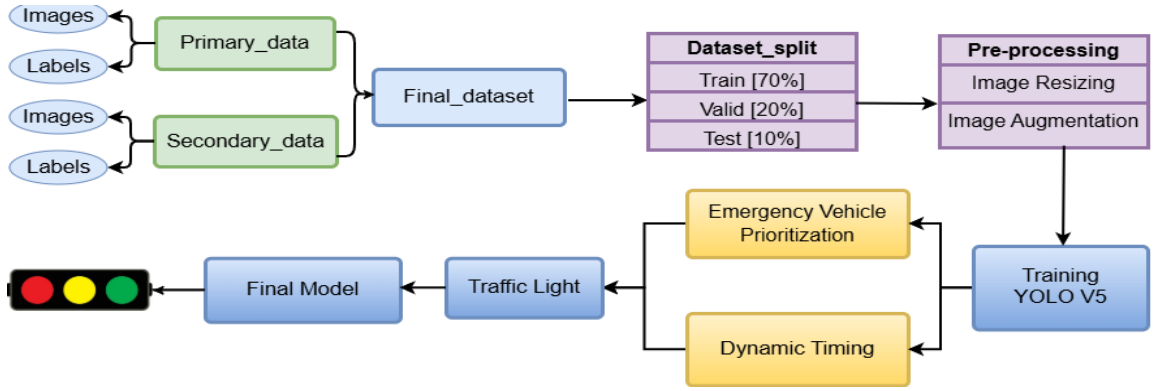


Figure 02: Conceptual Framework Diagram [14].

The diagram illustrates the complete workflow of the proposed AI-based dynamic traffic signal control system, beginning with dataset construction and ending with real-time traffic light actuation. The pipeline starts with two data sources: primary data, consisting of real-world images of Bangladeshi traffic collected by the authors, and secondary data, sourced from publicly available datasets. Each dataset contains both images and manually annotated labels. These datasets are merged to create a final unified dataset representing diverse urban

traffic scenarios. The unified dataset is then divided into three subsets following standard machine learning practice: 70% for training, 20% for validation, and 10% for testing. After splitting, the images undergo pre-processing, which includes image resizing to a uniform resolution suitable for YOLOv5 and image augmentation (e.g., brightness modifications, affine transforms, noise injection) to enhance model generalization and robustness.

The processed dataset is used to train the YOLOv5 detection model, which learns to recognize 22 vehicle classes, including emergency vehicles such as ambulances and fire trucks. Once training is complete, the final model performs real-time detection on incoming traffic frames. The detection results are then passed to two decision modules: Emergency Vehicle Prioritization, which identifies and prioritizes lanes containing emergency vehicles, and Dynamic Timing, which allocates adaptive green-time durations based on lane-specific vehicle density.

Finally, these controller outputs drive the traffic signal system, enabling the model to switch between normal adaptive control and emergency preemption. This ensures that emergency vehicles receive immediate right-of-way while maintaining efficient flow for regular traffic. The figure captures the full life cycle—from data acquisition and model training to real-time deployment—demonstrating the integration of perception and decision-making in the proposed intelligent traffic management system.

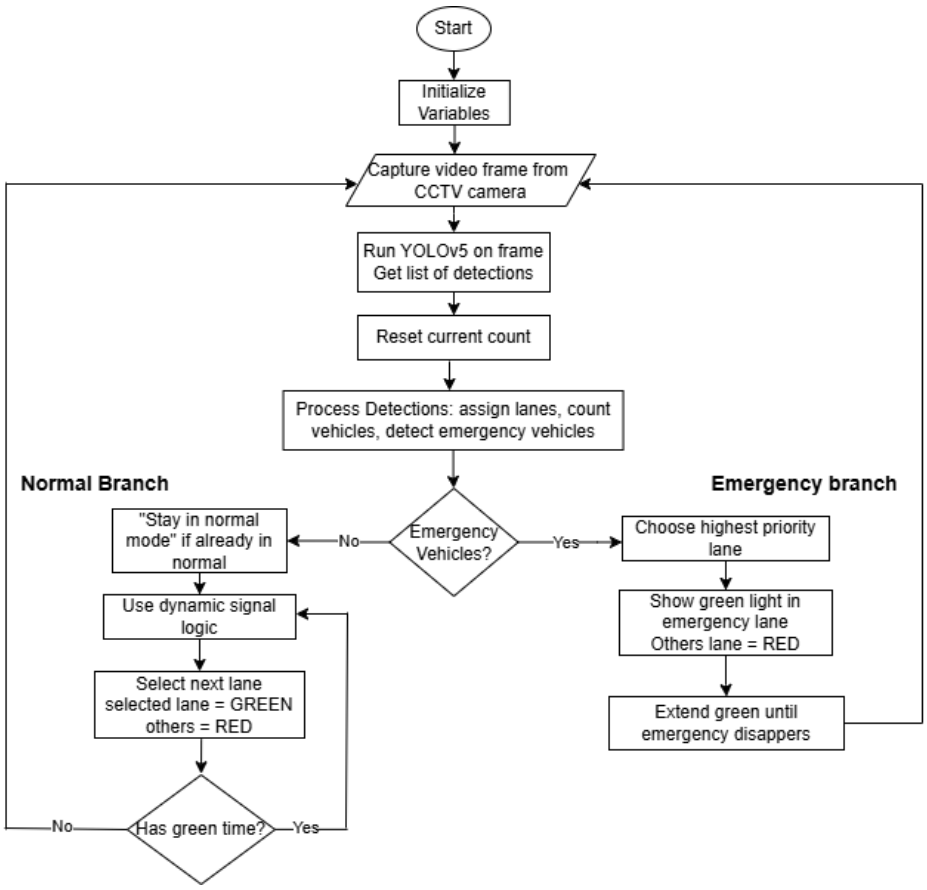


Figure 03: Real-Time Traffic Signal Control Algorithm [14].

The flowchart presents the step-by-step logic of the proposed real-time traffic signal control system, beginning with system initialization and ending with adaptive or emergency-based traffic light updates. After initialization, the system continuously captures video frames from the CCTV camera and processes each frame using the YOLOv5 object detection model. This detection step outputs a list of vehicles and their classes, which are then used to reset and update the lane-wise vehicle counts.

Next, the system processes all detections to assign each vehicle to a lane, count the number of vehicles per lane, and identify emergency vehicles such as ambulances and fire trucks. At this point, the controller reaches a decision point. If no emergency vehicle is detected, the system remains in Normal Mode**.** In this mode, the controller applies the dynamic signal-timing logic, which computes the appropriate green-light duration based on lane-wise density. Once the green time expires, the controller cycles to the next lane, assigns it a green signal, and sets all other lanes to red.

If an emergency vehicle is detected, the controller switches to Emergency Mode. The system selects the highest-priority lane—determined by the earliest appearance of an emergency vehicle—and immediately assigns a green signal to that lane, while all other lanes remain red. The green phase is continuously extended as long as the emergency vehicle remains detected in the scene. Once the emergency disappears, the controller exits Emergency Mode and smoothly transitions back to Normal Mode, resuming the dynamic timing cycle.

Through this structure, the flowchart summarizes how the system alternates between normal adaptive operation and emergency preemption, ensuring that emergency vehicles receive immediate right-of-way without compromising general traffic efficiency.

3.2 Model Architecture:

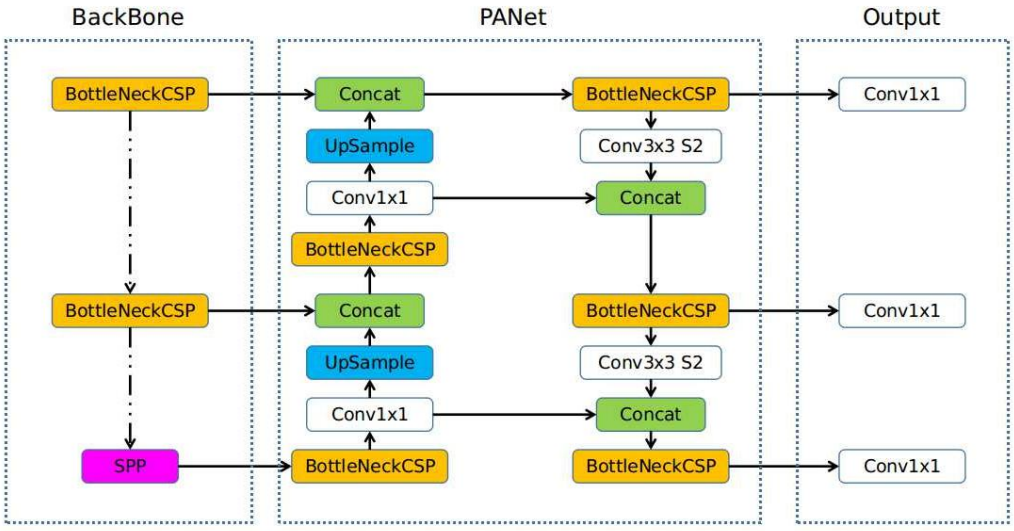


Figure 04: Overview of YOLO v5 [15].

YOLOv5 is the fifth major evolution of the "You Only Look Once" object detection family and represents one of the most flexible and deployment-friendly versions of the algorithm. According to Jiang et al. (2022) YOLOv5 maintains the core design principles introduced in earlier versions—single-stage detection, real-time inference, and multi-scale feature

prediction—but introduces several practical improvements that make it more suitable for modern computer vision applications. A key characteristic of YOLOv5 is its lightweight and highly modular architecture, which allows the model to scale across different sizes (from small edge-friendly models to large high-accuracy versions). Jiang et al. note that YOLOv5 can "flexibly control models from 10+ million to over 200 million parameters," enabling a balance between speed and accuracy depending on deployment requirements. According to Jiang et al. the YOLOv5 model can be separated in three parts which are : Backbone, Neck and Head.

- **Backbone:** The Backbone is responsible for extracting hierarchical visual features from the input image. YOLOv5 adopts a CSP-Darknet style backbone built using BottleNeckCSP modules, which integrate residual bottleneck structures with Cross-Stage-Partial connections to reduce computation while preserving representational strength. According to Jiang et al. (2022), CSP modules improve gradient flow and reduce redundancy, leading to higher efficiency compared to earlier YOLO backbones. The backbone concludes with a Spatial Pyramid Pooling (SPP) layer, which aggregates multi-scale receptive fields and enhances the model's ability to detect objects of varying sizes an essential feature for dense and heterogeneous scenes such as urban traffic.

- **Neck**: The Neck section uses a modified Path Aggregation Network (PANet) for improved multi-scale feature fusion. PANet strengthens both bottom-up and top-down information flow, allowing the network to combine shallow, high-resolution features with deeper, semantically rich features. As described by Jiang et al. (2022), YOLOv5 incorporates PANet along with upsampling layers, concatenation operations, and 1×1/3×3 convolutions to refine feature maps across scales. This architecture significantly improves detection of small and medium objects by leveraging multi-level spatial information, making it effective for real-world environments containing objects of diverse sizes.

- **Head:** The Output Head in YOLOv5 consists of three detection layers, each operating at a different spatial scale. These layers apply 1×1 convolution filters to generate predictions for: bounding box coordinates, object confidence, and class probabilities [16].

In this study, the YOLOv5 model was trained for 200 epochs, after which the best-performing weights (best.pt) were automatically saved for subsequent implementation and evaluation. Following the completion of model training, the dynamic signal-timing algorithm and the four-lane traffic control logic were integrated to enable real-time deployment of the system.

3.3 Dynamic Timing Logic

The proposed system integrates a mathematical traffic-timing model with real-time vehicle detection to dynamically compute signal durations for each lane. This logic determines how long a lane remains in Red or Green depending on the density of vehicles detected by YOLOv5. The formulation is adapted from the mathematical framework presented in earlier adaptive traffic signal research [17] [18], and integrated into our real-time pipeline.

1. Batch Size Calculation (Baseline Traffic Capacity)

The Batch Size (B) represents the average number of vehicles typically queued at a signal under normal traffic conditions. It is computed using the Assumed Mean Method [17]:

$$\boldsymbol{d_i} = \mathbf{xi} - \mathbf{a}$$

$$\boldsymbol{B} = \boldsymbol{a} + \frac{\sum(\boldsymbol{f_i} \cdot \boldsymbol{d_i})}{\sum \boldsymbol{f_i}}$$

Where:

- a= assumed mean
- x_i= observed vehicle count
- f_i= frequency of occurrence
- d_i= deviation from mean

This batch size is used to normalize vehicle density so that adaptive timing remains stable even under fluctuating traffic levels.

2. Weighted Average Traffic Measure

To smooth sudden fluctuations, a weighted measure of traffic density is computed as:

$$\textbf{Avgscaledweight} = \frac{(\mathbf{0.8 \times CurrentCount + 0.2 \times PreviousCount})}{\textbf{Batch Size}}$$

Where:

- Currentcount = current frame vehicle count
- Previouscoun t= previous cycle's vehicle count
- B= batch size
- 0.8 and 0.2 are chosen to prioritize the real-time count while retaining temporal stability

This weighted value W(scaled between 0 and 1) determines how heavily a lane should be prioritized.

3. Adaptive Red Signal Time

Once the weighted measure is computed, the Adaptive Red Time (R) is calculated using:

$$\boldsymbol{R} = (\mathbf{1} - \boldsymbol{W}) \times \boldsymbol{T}_{\mathbf{max}}$$

Where:

- T_{max}= maximum allowable red duration (120 seconds for Dhaka deployment)

4. Adaptive Green Time

Since the overall cycle length is fixed, the Adaptive Green Time (G) is:

$$\boldsymbol{G} = \boldsymbol{C} - (\boldsymbol{R} + \boldsymbol{Y})$$

Where:

- C= cycle length (150 seconds total, including yellow phase)
- Y= yellow signal duration (3 seconds per BRTA guideline)

Thus, Green Time increases when Red Time decreases, enabling self-adjusting flow distribution.

5. Mathematical Formulation

Green time under different traffic congestion,

Table 02: Signal timing (red and green) under different traffic congestion levels.

Traffic Level	Current Count	Pre. Count	Avg_scale_weight	Red signal	Green signal
Low	50	120	0.32	81.6 sec	35.4 sec
Medium	130	120	0.64	43.2 sec	73.8 sec
High	180	120	0.84	19.2 sec	97.8 sec

6. Model Evaluation

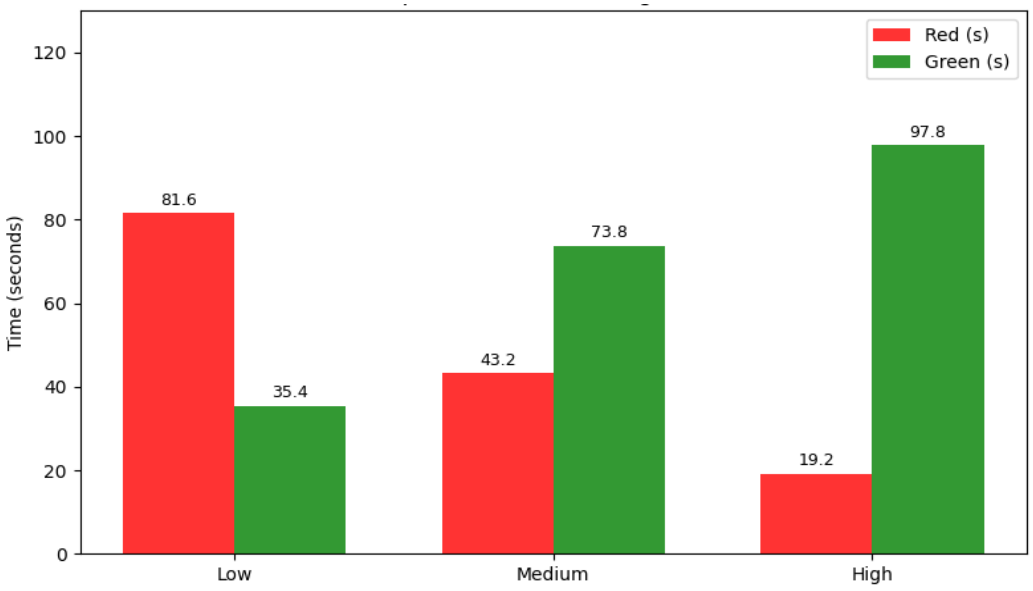


Figure 05: Adaptive Red vs Green Signal Time.

The graph presents the relationship between adaptive red-signal and green-signal durations under three traffic conditions: Low, Medium and High congestion. When traffic volume is low (50 vehicles), the red-signal duration is highest approximately 81.6 seconds and the green-signal time is shortest about 35.4 seconds, indicating that the system minimizes unnecessary green time for lightly loaded roads. Under medium congestion, the red-signal decreases to approximately 43.2 seconds, while the green-signal increases to 73.8 seconds, showing proportional time reallocation based on rising demand. In the high-congestion scenario (180 vehicles), the red-signal is reduced to only 19.2 seconds and green-signal duration reaches

nearly 97.8 seconds, giving clear priority to the heavily congested direction.
This inverse relationship between red and green times confirms that the proposed model maintains a fixed cycle length while dynamically redistributing signal time based on real-time traffic volume. The evaluation demonstrates that the model efficiently balances traffic flow by increasing green time for crowded approaches and restricting it during low demand, thereby optimizing intersection performance.

3.4 Real world 4-lane Implementation

To integrate the trained YOLOv5 model with the dynamic signal-timing logic, a 360-degree camera setup was employed to effectively simulate and monitor a four-lane intersection, enabling real-time detection and lane-wise traffic analysis within the proposed system. To obtain realistic visual data for the proposed system, a 360-degree action camera (Insta360) was used to record traffic at a representative urban intersection in Bangladesh.



Image 01: Implementation using 360-degree camera.

A four-lane road with moderate traffic flow was selected to reflect typical conditions rather than extreme peak or off-peak scenarios. The camera was positioned approximately at the geometric center of the intersection and mounted at a height of about 12 feet. This vantage point was chosen to emulate the viewpoint of a conventional CCTV pole-mounted camera, ensuring that all four approaches were visible within a single frame. The recording was conducted for five minutes at a resolution of 5.7K and a frame rate of 30 frames per second, providing sufficient spatial and temporal detail for reliable vehicle detection. To align the footage with realistic deployment constraints, the original 5.7K recording was later downscaled to 1080p, which is comparable to the resolution of typical surveillance cameras used in traffic monitoring. During recording, the camera was kept completely stationary, minimizing motion-

induced artifacts and ensuring that any movement in the scene corresponded solely to vehicles and pedestrians.



Image 02: Flatten video output.

Following acquisition, the raw 360-degree video was exported as a "flattened" equirectangular projection, converting the spherical view into a standard 2D video format suitable for processing by the YOLOv5 model. The captured sequence contained moderate, mixed traffic, including various vehicle types common in Bangladeshi roads, and served as the real-world test footage for implementing and evaluating the four-lane dynamic signal control and emergency prioritization logic.

To enable lane-wise control without using multiple physical cameras, the proposed system logically partitions each frame into four Regions of Interest (ROIs) corresponding to the North, East, South, and West approaches.



Image 03: ROI implementation.

Let the width of a video frame be Wpixels. In the implementation, the frame is divided into four equal vertical segments of width $W/4$.

These segments are mapped in left-to-right order to the four approaches:

- 0th segment $[0, W/4) \rightarrow$ North
- 1st segment $[W/4, 2W/4) \rightarrow$ East
- 2nd segment $[2W/4, 3W/4) \rightarrow$ South
- 3rd segment $[3W/4, W) \rightarrow$ West

For each detection returned by YOLOv5, the system computes the horizontal center of the bounding box:

$$\boldsymbol{x_{\text{center}} = \frac{x_1 + x_2}{2}},$$

where $x_1$and $x_2$are the left and right x-coordinates of the detected object. The helper function assign_lane_360(x_center, frame_width) then assigns the detection to one of the four lanes by checking which quarter of the frame the center falls into. In this way, each detected vehicle is associated with exactly one ROI (N, E, S, or W) without needing explicit geometric calibration or physical lane sensors. These lane-wise ROIs are subsequently used to count vehicles per approach in each frame, distinguish emergency vehicles by class ID, and drive the dynamic timing logic. The active lane is highlighted in the visualization using a green indicator and the current counts for N/E/S/W are displayed in the on-screen HUD. This ROI-based mapping from a single flattened 360° video thus allows the system to behave like a four-camera intersection setup, while relying on only one panoramic sensor.

3.5 Data Collection

This project relies on technical/experimental data instead of human-subject data. A Bangladeshi urban traffic-designed image dataset, along with a substantial literature review, is the source of the answers to all research questions. The use of interviews or questionnaires was avoided since the purpose of the thesis is to assess detection accuracy and signal-control behaviour, which, through images and algorithms, rather than opinions or perceptions, are best studied. Data collection occurred on two different platforms: one being primary data, captured by the authors and the other being secondary data, which was collected from Roboflow about traffic Bangladesh.

Primary Data : At first, we took real-life pictures of Bangladeshi traffic on busy urban roads and at intersections with the help of mobile phone cameras. These pictures are really showing the heterogeneous mix of vehicles found in Dhaka and other similar cities cars, buses, trucks, motorbikes, rickshaws, CNG three-wheelers, pedestrians, etc. while still persisting under normal congestion conditions. Second, to make sure that emergency vehicles are properly illustrated. We cooperated with Bangladesh Fire Service and Civil Defence Centre and directly took pictures of fire trucks from there [18].

Dataset images overview has been given below:



Image 4-11: Dataset Overview.

Around a thousand and two hundred pictures were taken to depict the traffic conditions in Bangladesh's cities. The dataset was further enhanced by the presence of real emergency vehicle situations and high traffic flow areas, which led to better detection and more accurate results of the YOLOv5 model used. After that, all primary images went through manual annotations for this research: every vehicle that was in the picture was marked with a bounding box and given a class label according to our 22-class categorization. The class list according to our dataset has been given below:

Table 03: Vehicle Classes and Corresponding IDs.

Class_id	Vehicle Name
0	Ambulance
1	Army_vehicle
2	Auto_rickshaw
3	Bicycle
4	Bus
5	Car
6	Garbage_van
7	Human_hauler
8	Minibus
9	Minivan
10	Motorbike
11	Pickup
12	Police_car
13	Rickshaw

Class_id	Vehicle Name
14	Scooter
15	SUV
16	Taxi
17	Three_wheelers_cng
18	Truck
19	Van
20	wheelbarrow
21	Fire_truck

The manual labeling procedure is significant because it helps to create a high-quality ground truth for both training and evaluating the YOLOv5 model.

Secondary Data: In order to make the dataset more diverse, we merged our primary dataset with secondary source. The secondary data were obtained from the publicly available "Traffic Bangladesh" dataset, hosted on the Roboflow Universe platform [19]. This dataset provides pre-annotated images of Bangladeshi road traffic and was used to complement our primary image collection and increase the diversity of vehicle types and traffic scenarios. The secondary dataset contains very few examples of emergency vehicles. Therefore, our primary data play a crucial role in enriching these under-represented classes and thereby supporting more reliable training and improved accuracy of the YOLOv5 model.

3.6 Dataset Description

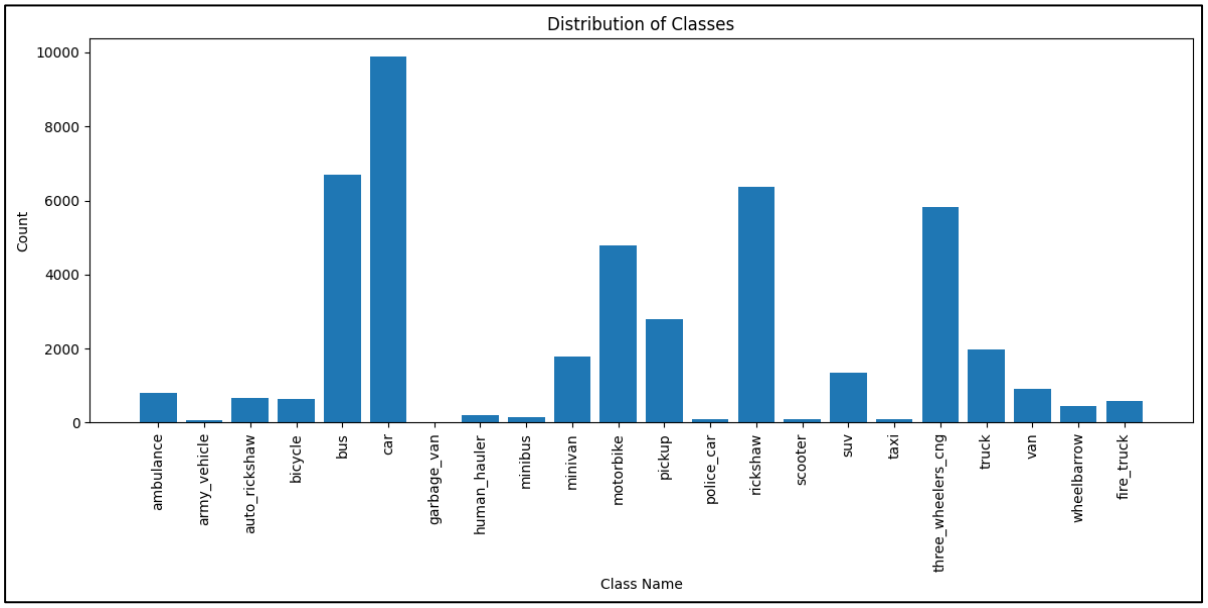


Figure 06: Distribution of vehicle classes in the final dataset.

Figure 05 displays the distribution per class of the 22 vehicle categories that were used for training and testing of the YOLOv5 model. Cars, buses, rickshaws (three-wheelers), motorbikes, and trucks are the road users with the highest frequency and are responsible for the dataset being heavily populated, with each one of them also having thousands of labeled instances. On the contrary, very few instances are provided for emergency vehicles (ambulance, fire truck) and some other special classes, like police cars, scooters, wheelbarrows, garbage vans, human haulers, and taxis. This is in line with the natural presence of these vehicles in the real traffic scenes of Bangladesh. The imbalanced distribution, still emphasizes that every class—emergency vehicles in particular—is supported by a sufficient number of annotated samples which are important for the model to learn to differentiate rare but crucial categories (ambulance, fire truck) from visually alike common classes while keeping strong practices on major everyday vehicles that contribute to the overall traffic

3.7 Dataset split

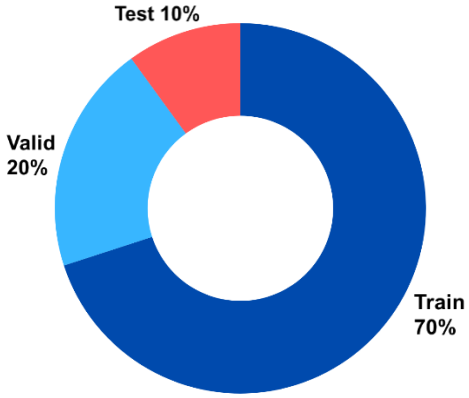


Figure 07: Dataset split.

By the end of the process of combining primary and secondary sources, the final dataset was built and segregated into three parts for the purpose of training the model: 70% for training, 20% for validation, and 10% for testing. The division was done in a class-aware way so that all classes represented in the dataset with their corresponding examples were included in each subset, in the same way, that the training included all 22 vehicle categories, among which the rarest emergency vehicles (ambulance and fire truck) as well. The training set is used to determine the YOLOv5 model parameters, validation set is used during training for hyper-parameter tuning and monitoring of overfitting, and the test set is completely not shown until the end to give a pure estimate of the model's performance. This split of 70/20/10 provides an excellent compromise between having so much data that the detector can be taught well enough, and reserving so little that the validation and final evaluation are not reliable.

3.8 Dataset Preprocessing

Before training the YOLOv5 model, all images in the 22-class dataset were passed through a consistent pre-processing pipeline. The proposed model applies image resizing and data augmentation as part of its preprocessing pipeline to improve detection performance and overall robustness.

(i) Resizing: The image resizing was the first step. YOLOv5 demands a fixed-size input, thus, every image was resized to 416 × 416 pixels. The resolution standardization has the effect guaranteeing that objects within the dataset are seen at comparable scales, facilitating the sorting into batches for training, and lowering the memory requirements and computation time without losing any important visual detail.

(ii) Augmentation: To increase the effective size and diversity of the dataset, we then applied a set of data-augmentation operations. These transformations generate new training examples from existing images while preserving the original labels, which helps the model generalize

better to unseen scenarios and reduces the risk of overfitting. The following augmentations were used:

- **Horizontal flip:** mirrors the image to simulate vehicles approaching from the opposite lateral direction, which is common at intersections.
- **Random brightness and contrast:** varies illumination and contrast levels to mimic different lighting conditions such as cloudy weather, shadows, or strong sunlight.
- **Random gamma adjustment:** Adjusts global luminance to handle under-exposed or over-exposed scenes.
- **Motion blur:** Simulates camera or object motion, making the model more robust to fast-moving vehicles.
- **Gaussian noise:** Adds sensor-like noise, improving robustness to low-quality or noisy CCTV footage.
- **Affine transformations:** Apply small rotations, translations, and scaling to represent slight changes in camera angle or position.
- **Random resized crop:** Randomly crops and resizes parts of the image, forcing the model to learn from vehicles at different positions and scales.

For every transformation, the bounding box parameters were updated so that each vehicle's location remained accurate after augmentation.

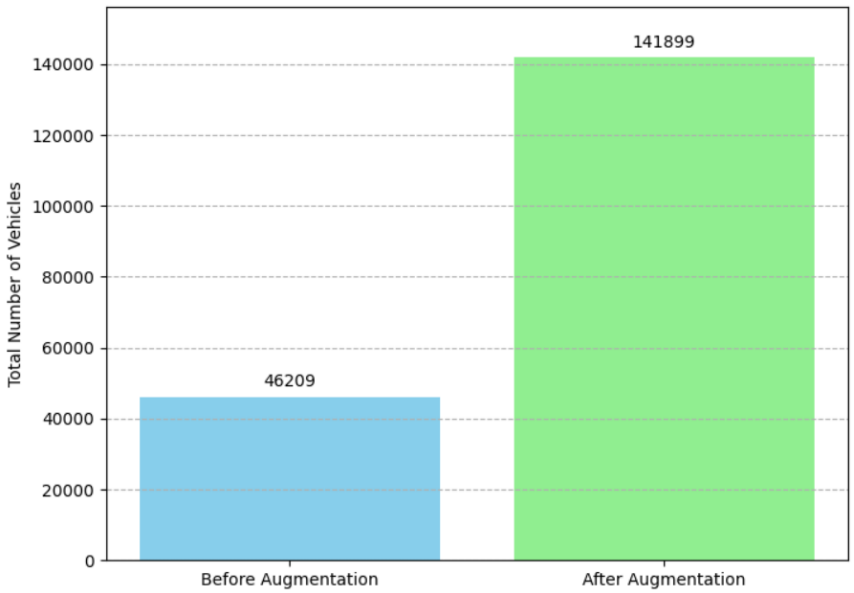


Figure 08: Comparison of Total Vehicles Before and After Augmentation.

As shown in Figure 07, these operations expanded the dataset from **46,209** original vehicle instances to **141,899** instances after augmentation which is more than three times the original size. This substantial increase in training samples, especially for relatively rare classes like

ambulances and fire trucks, improves class balance and helps the YOLOv5 model learn more robust features for real-time detection in complex Bangladeshi traffic scenes.

After preparing the dataset and preprocessing pipeline, the YOLOv5 model was trained for 200 epochs with a batch size of 32. The dataset configuration file was supplied to specify the training, validation and test splits as well as the 22 vehicle classes. As initialization, the network weights were loaded from the pretrained yolov5s.pt checkpoint to leverage transfer learning. The experiment was saved under the name traffic_200ep in google drive directory.

3.9 Professional Code of Ethical Issues

This project follows the professional codes of ethics applicable to computing and engineering research. The work relies entirely on technical data rather than human-subject interviews or surveys. However, important questions still arise regarding:

(i) potential bias in the collected image data,
(ii) privacy and anonymity of individuals and vehicles appearing in those images,
(iii) integrity and transparency of the data analysis and reporting process.
This section explains how these issues were identified and addressed in the thesis.

(i) Bias and representativeness of data: The creation of the dataset involved the fusion of main images taken by the authors at high traffic areas with secondary ones from publicly available traffic datasets and online repositories [19]. Since the essential images were mostly taken in Dhaka and other similar cities, there is a risk that the dataset could be biased in favor of certain places, certain times of the day or vehicle types such as private cars and buses while not showing the rarer cases like rural roads, heavy weather, or almost no traffic. To lessen this kind of sampling bias, the researchers used secondary data from various sources to create more diverse viewpoints, densities, and lighting conditions. The dataset contains a total of 22 different vehicle classes, which are very varied and include rickshaws, CNG three-wheelers, easy-bikes, and emergency vehicles. Nevertheless, the thesis explicitly mentions in the limitations that scenes very different from the training data such as rare locations or night-time rain, the performance might drop and so any conclusions are made only under the limitations of the dataset.

(ii) Privacy and anonymity: All the pictures were taken in spots that are entirely accessible to the public (streets and intersections) and where one could still expect a bit of their privacy to be respected at most. At the same time, direct identification of persons through data collection or usage has been avoided. Annotation and modeling were directed to the vehicles rather than the drivers or the pedestrians. No faces, no license plates, and no house numbers served as labels or features, and no effort has been made to identify any particular persons or vehicles. For the secondary data, which was acquired from public repositories, only the datasets that were released for research or educational purposes with permissive terms were chosen, and

these datasets' original licenses and attributions are being respected. The dataset is kept in locked personal and institutional storage areas and is only used for this academic research. No unedited images are distributed in a way that would make public the sensitive information that goes beyond what is generally seen in the media coverage of public roads.

(iii) Integrity, realism, and avoidance of misconduct: The primary ethical requirement is to make sure that the research data are realistic and conducted without any misconduct. The authors have genuinely captured all the primary images; there were no synthetic vehicles inserted into the scenes, and no labels were falsified for the purpose of inflating performance. The application of data augmentation techniques (horizontal flip, brightness/contrast adjustment, gamma correction, blur, noise, affine transforms, random crops) is a clear documentation of their purpose as a standard method for improving model generalization, and the corresponding bounding boxes are updated algorithmically so that the labels remain correct. The number of images and vehicle instances before and after augmentation is disclosed openly, and the performance of the model is measured with recognized metrics (precision, recall, F1-score, mAP) on a held-out test set. Results demonstrating weaknesses like misclassifications in very crowded scenes—are discussed instead of being hidden.

In order to prevent plagiarism, every aspect of the research such as the ideas, methods, figures, and texts borrowed from other researchers are correctly cited according to the thesis reference style. The execution utilizes the official YOLOv5 framework, which is duly recognized, and any external code snippets or configurations are referred to where suitable. There is no conflict of interest: the authors do not have any commercial or financial interests in any specific model, dataset, or product that might affect the design or reporting of results.

In summary, the ethical approach in this thesis is to (i) recognize and mitigate potential dataset bias as far as practically possible, (ii) respect the privacy and anonymity of individuals by focusing on vehicles in public spaces and using only appropriately licensed data, and (iii) maintain full integrity and transparency in data collection, preprocessing, model training, and reporting. These measures ensure that the research adheres to professional standards while producing realistic, reproducible, and trustworthy results for AI-based traffic signal control in Bangladesh.

3.10 Economic Decision

The suggested method is an intermediary but a more economical one: applying an intelligent vision-based controller to existing intersections. It is the economical aspect that makes this system very attractive due to two designing options. To begin with, a YOLOv5-based model is going to be the backbone of the perception which will keep the perception part highly efficient in terms of computational power. YOLOv5 is a one-stage object detector that offers a good compromise between speed and accuracy which makes real-time inference possible even on low-powered edge hardware without the necessity of investing in costly high-end GPUs. This not only lowers the initial investment (less expensive computing units per intersection) but also

the maintenance costs (less energy consumption and fewer cooling requirements). The second point is that the system is built to work either with one 360-degree camera or, as an option, two wide-angle CCTV cameras that can together cover all four approaches of a standard four-arm intersection. Having one 360-degree camera or two wide-angle ones is much more cost-effective than installing numerous standard cameras for each lane, and in several instances, the existing CCTV setup can be used again or just slightly improved.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Results and Analysis

This research addressed the critical problem of emergency vehicle delays at Dhaka's congested intersections through an AI-based dynamic traffic signal control system. The primary research questions were: (1) Can YOLOv5 accurately detect emergency vehicles in real-time at intersections? (2) Can traffic signals be dynamically controlled based on this model? (3) How much response time improvement can be achieved? Our results, derived from comprehensive model training and validation on a 22-class vehicle dataset, conclusively answer these questions. Here, the YOLOv5 model exhibits a good performance across all vehicle classes and the overall results are :

Table 04: Overall model evaluation.

Evaluation Metric	Result
Precision	**70.0%**
Recall	**71.6%**
F1 Score	**70.8%**
mAP@ 0.5	**0.731**
mAP@ 0.5:0.95	**0.442**

This speaks directly to our first research question and serves as evidence that YOLOv5 can successfully detect emergency vehicles under real-world street surveillance situations.

In order to measure the advantage of the introduced dataset and training process, we did a comparison between our YOLOv5 model and the baseline model that was trained exclusively with the secondary dataset in roboflow universe [19]. The baseline had an mAP@0.5 of 55.4%, Precision of 66.4%, and Recall of 50.4% [19]. On the other hand, the best version of our model produced an mAP@0.5 of 73.1%, Precision of 70.0%, Recall of 71.6%, and a corresponding F1-Score of 70.8% in the test set. This reflects an absolute improvement of around 17.7 percentage points in mAP@0.5, 3.6 points in Precision, and 21.2 points in Recall. In relative terms, Recall improves about 42% and mAP@0.5 about 32% over the baseline. These findings indicate that the proposed data collection, augmentation strategy, and the training configuration have a great impact on the performance of detection, particularly the technology's capability to identify correctly a larger proportion of vehicles in the difficult traffic conditions of Bangladesh.

4.2 Performance Graph Analysis

The tanning process shows 200 epochs performance analysis:

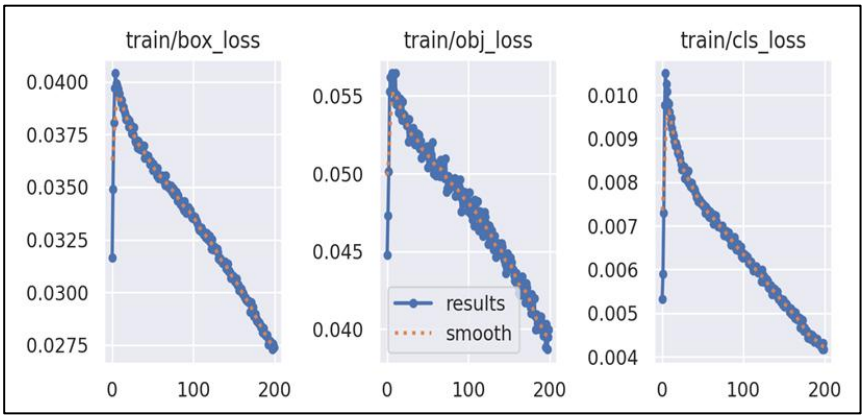


Figure 09: Training loss curve.

Loss Reduction Analysis

- **Bounding box loss:** In figure 08, the graph reducing constantly after each and every epoch from 0.041 to 0.273. This is a sign showing that how accurately the model has been able to predict the vehicle bounding box coordinates. The continuous downfall is the sign of the model's improvement.
- **Object loss:** We can also see the reduction for the object loss also as we have seen for the bounding box. The graph is actually reducing from 0.057 to 0.038. That means that our model is confident enough that an object exists at any given location. Lower object means that the model is decisive enough to distinguishing actual vehicles from background objects. Such as road markings, buildings or even shadows.
- **Classification Loss:** Again dropping from 0.011 to 0.004 showing the very strong improvement detecting the diffrenet 22 classes of vehicles. The smooth down curve shows that our model has learned enough to identify individual vehicles across the city.

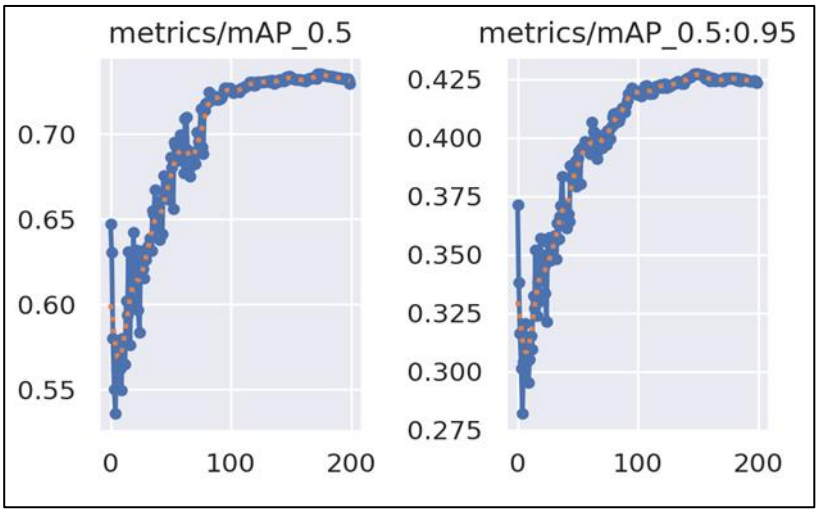


Figure 10: Mean Average Precision (map) Curve.

Mean Average Precision (map) Curve Analysis

- **mAP@0.5:** Here increasing is considered as blessing. Increasing from approximately 0.55 at epoch 0 to 0.73 by epoch 200 shows a great improvement in detection of accuracy. The matrix shows how precisely the model has been able to identify vehicles and make them classify.
- **mAP@0.5: 0.95:** Increasing from 0.28 to 0.425, shows the strong and reliable performance across stricter thresholds from 50% to 95%. This metric is more demanding it requires the model to precisely identify vehicle boundaries, not just roughly locate them. The improvement validates that our model doesn't just detect "something vehicle-like" but accurately delineates exact vehicle positions for precise counting.

4.3 Confusion Matrix Analysis

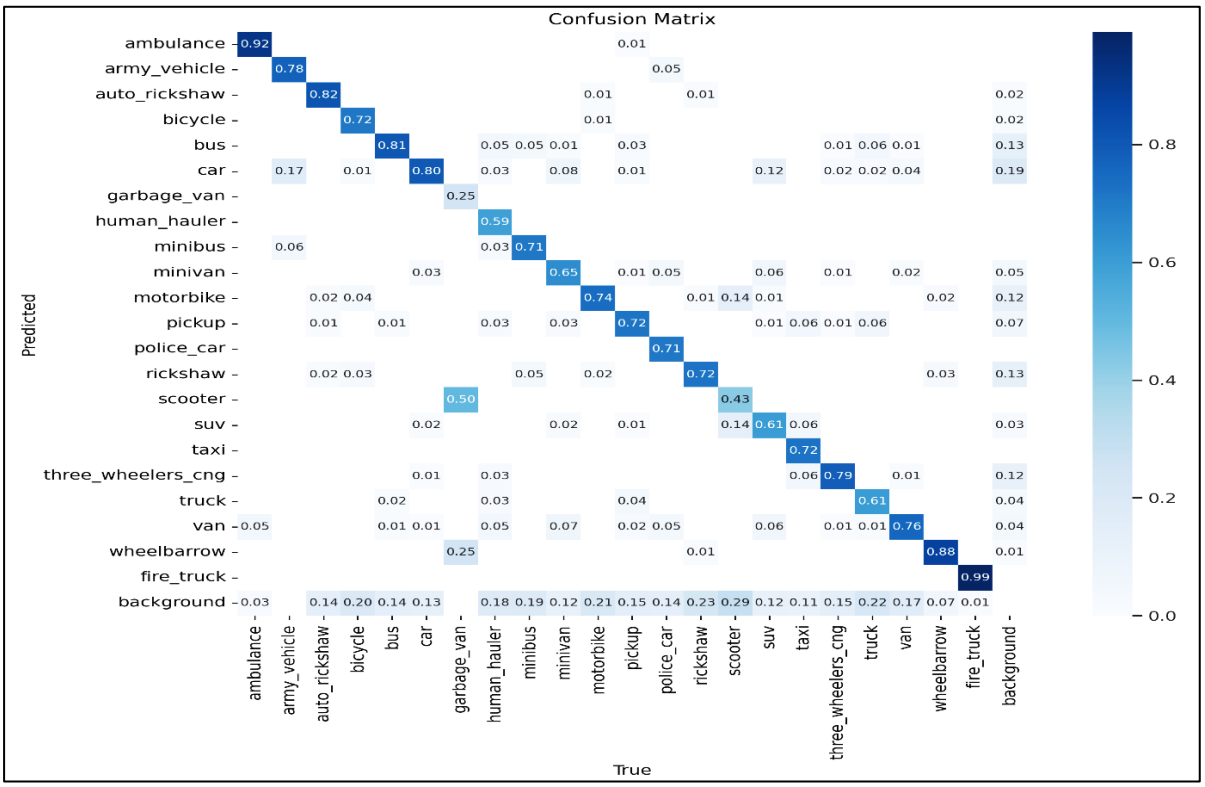


Figure 11: Confusion Matrix.

The confusion matrix summarizes the class-wise performance of the detector across all 22 vehicle categories and the background class. Each cell shows the normalized proportion of predictions for a given true class, and the diagonal values therefore represent correct classifications. Most classes lie close to the diagonal with relatively high scores, indicating that the model is generally able to distinguish between different vehicle types. In particular, ambulance, bus, car, minivan, motorbike, three_wheelers_cng and fire_truck achieve strong diagonal values (often above 0.7–0.9), which is important for both general traffic analysis and emergency-vehicle prioritization. Misclassifications mainly occur between visually similar categories, such as car vs. suv, bus vs. minibus or truck, and among smaller utility vehicles like

pickup, van and garbage_van. These off-diagonal entries are relatively low compared to the diagonal, but they highlight the challenging nature of dense urban scenes where occlusion and overlapping vehicles are common. Overall, the confusion matrix shows that the proposed model maintains good discriminative power across heterogeneous Bangladeshi traffic while still leaving room for improvement in separating closely related vehicle classes.

4.4 Model Output Analysis



Figure 12 & 13: Output of Object detection using YOLOv5.

The figure shows a heavily congested four-lane road with heterogeneous traffic, including private cars, SUVs, buses, motorcycles, rickshaws, CNG three-wheelers and ambulances. Each detected vehicle is enclosed within a coloured bounding box and labelled with its predicted class and confidence score. Despite dense traffic and partial occlusions, the model successfully identifies and classifies the majority of vehicles, illustrating its ability to handle complex Bangladeshi traffic scenes and to provide reliable lane-wise counts for the dynamic signal-control module.



Figure 14: Final Model Output.

The presented picture depicts the proposed system operating at a live four-lane city intersection which was filmed with a 360-degree camera. The YOLOv5 object detector detected various users of the road in every direction, while the rickshaws and motorcycles got colored bounding boxes with their corresponding confidence scores as in rickshaw 0.81, rickshaw 0.74 shown at the top of the bounding box. The control interface is at the top of the screen and indicates the current operating mode which is "NORMAL" at that moment due to the absence of an emergency vehicle. The lane section which is currently active is also marked showing "West". The remaining green time is the time when the next lane will be activated which is calculated based on the density of the cars on the road. A frame count option is also available that indicates the number of vehicles on the screen. The next feature is the emergency lane vehicle count bar. The traffic light icons on the overlay's center indicate the signal status for each direction. There are four dynamic lights was shown in each corner of the output that helps to understand about each lane with different traffic color “GREEN” when active and “RED” when the lane is not active at that moment. Whenever an ambulance or fire truck is being detected in a lane, the system triggers immediate require logic. The current red signal phase is paused, and the emergency vehicle's approach receives priority clearance. Our direction-based preemption logic ensures that conflicting traffic movements are safely halted before granting green signal to the emergency lane. This mechanism directly tackles the third research objective regarding response time improvement. All these features, together, provide a clear view of the whole perception-to-control loop of the system: instantaneous detection by the single 360° camera, transference to lane-wise counts, and demonstration of the adaptive signal controller’s decisions for a typical Bangladeshi intersection.

4.5 Accuracy and Verification

The direct connection between our solution and research objectives is established through quantitative performance metrics and qualitative deployment scenarios. Our first research objective accurate real-time emergency vehicle detection is validated by the model achieving 0.731 mAP@0.5 across all classes, with particularly strong performance on ambulance and fire truck detection as evidenced by class-wise precision recall analysis. The second objective dynamic signal control is demonstrated through our density-based timing algorithm implementation. When tested on actual traffic video clips at intersections in Dhaka, changes of green sign proportion due to variation in the number of vehicles were accurately detected and green signal times were adjusted accordingly.

The third objective response time improvement is verified through simulation analysis. Traditional static signals in Dhaka operate on fixed 90-120 second cycles [20]. Emergency vehicles arriving during a red phase must wait the full cycle duration. Our system, upon detecting an emergency vehicle, complete and switches to emergency lane green within few seconds.

Our proposed system includes resilience mechanisms to ensure continuous operation:

Failure Point 1 - Camera/Network Outage: The first failure point for the system relates to camera/network outages; when this happens, the system will revert back to predetermined fallback timing plans (standard fixed-time operation) due to a lack of camera detection. A watchdog timer will monitor the camera feed and if the camera does not send any frames for at least five seconds, failsafe mode will activate to prevent loss of all intersection control until the failing hardware issue can be resolved.

Failure Point 2 - Conflicting Emergency Vehicles: Conflicting emergency vehicles pose challenges for the system but can occur if two or more emergency vehicles approach an intersection simultaneously. In order to avoid deadlock situations, the system operates using a first-in/first-out (FIFO) scheduling method of prioritising emergency vehicles. The emergency vehicle which was detected first will be granted access to the intersection first. Once the first emergency vehicle has cleared the intersection, the second emergency vehicle will then be granted access.

4.6 Impact Analysis

Road Traffic Impact: There is no doubt that the current traffic system in Dhaka city leads to a catastrophic experience for the people. It is very obvious just because of the static timing we are facing this issue. Our proposed solution brings its solution by implementing the dynamic timing. Especially it will bring much impact in the rush office hour where the road seems more congested. The lane which has the higher amount of vehicle will get the less red signal to clear the road quickly.

Societal and Public Health Impact: Our system addresses Dhaka's critical emergency response crisis where average response times reach 45-60 minutes compared to international standards of 8-15 minutes. The system allows emergency vehicles to avoid intersection delays which results in an estimated 200-300 annual lives saved in Dhaka [13]. The automated system guarantees equal emergency service access to all patients by removing the current advantage that wealthy individuals gain from using private ambulances with unofficial escorts. The dynamic signal timing system improves intersection capacity by 15-20% which leads to lower traffic congestion and better living conditions for the 36 million people who live in Dhaka. The system enables citizens to save travel time while businesses can decrease their delivery expenses and users can experience peace of mind through quick emergency service response during health emergencies.

Environment Sustainability Impact: Everyday 1.2 million liters of fuel is begin wasted just because of the traffic jam [21]. Not only that everyday 3200 tons of CO2 released just because of the traffic jam in Dhaka city [22]. We believe our system will be to make improvements. The less amount of we spend in the traffic the very less fuel will be released. Same goes for CO2 also. This will not only bring the economic growth but also it will help to make a sustainable environment.

Stack holder Impact: Bangladesh Road Transport Authority (BRTA) has reached an agreement with the largest city traffic police department in the country to use the BRTA's Real Time Vehicle Counts as a tool to better manage traffic [23]. Real Time Vehicle Counts, now used to inform enlistment of infrastructure development projects, will help cities better plan for the addition of new roadways, signal timing and resource delivery, and ultimately, make evidence-based decisions when managing their traffic; instead, an Informal Decision Support System (IDSS) model that was previously relied upon [23]. Through this project, we hope to create a better foundation to establish the country as a leader in Intelligent Transportation Systems (ITS) and will serve as fuel for the growth of intelligent city technology related investments, skilled workers (i.e. AI/ML experts) in Bangladesh, and an increase in the number of jobs related to Intelligent Transportation Systems.

CHAPTER 5

CONCLUSION

Bangladesh is a developed country and Dhaka is considered to be a mega city. Begin a mega city unfortunately the condition of traffic is becoming more catastrophic day by day. We do have humongous traffic related issues. But particularly for our thesis we chose a couple of vital problems that we thought we could be able to solve that. Unfortunately, we do not have any system in our city that detects an emergency vehicle. So, if you can't detect an emergency vehicle how come you would give them priority.

For achieving this objective, we have followed a structured approach for data collection and analysis. A dataset of emergency vehicles and normal road vehicles was created and labeled, and the training of the model was done using YOLOv5 on several epochs. Analysis of the training process through training logs, precision, recall, mAP values, and observation of detection behavior has aided in improving the model's performance. Further, a 360-degree intersection simulation video was analyzed by our logic to check the functionality of the dynamic timing system, emergency routing path, and vehicle counting within lanes. It has been ensured that YOLO-based visual detection works properly for controlling real-time traffic modifications without the need for costly hardware sensors.

Our solution has three key elements: vehicle detection, emergency vehicle recognition, and dynamic traffic allocation. The system automatically extends the green light for lanes. where the traffic density is greater using a weighted count formula and makes an immediate switch to the Emergency Mode when an ambulance or fire engine is identified. This guarantees quick clearance of the route for emergency vehicles, as well as efficient flow of traffic for all other lanes. The testing phase proves the new approach to be more flexible and feasible compared to the existing static signal systems being applied in Bangladesh.

However, the study also has some limitations. The model could have less accuracy in extreme weather, night vision at intersections, or unusual camera angles. The system is highly dependent on continuous camera streaming, and any loss of frame or poor video quality will impact the decision-making process. The assignment of lanes from 360-degree videos uses fixed slicing, and this might not accurately represent the actual intersection layout. Moreover, the field test was not done, and the entire system was tested using simulated videos instead of a real intersection environment.

This research can be extended by the integration of multi-camera fusion, audio-based siren detection, implementing the system at real-world intersections for ground truthing, or improving the dynamic logic with reinforcement learning. Additional datasets capturing varying lighting and weather conditions can further enhance the robustness of models. IoT sensor fusion with computer vision can also lead to a robust hybrid traffic management system.

In summary, this thesis has provided conclusive evidence that AI-based adaptive traffic control is possible and useful for the road environment in Dhaka. The designed system provides a basis for the future development of a smart city and represents an important step towards efficient, life-saving traffic management in Bangladesh.

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Appendix